

23 Pricing and financing

12 primary questions | Oldest to newest

2026 objectives: 4.3

11 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

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CA1-P1 September 2010 Q7	15 marks
CA1-P1 April 2011 Q1	8 marks
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CP1-P1 September 2023 Q4	12 marks
CP1-P1 September 2024 Q9	12 marks

CA1-P1 | September 2010 | Q7

Current topic | 15 marks total | Original PDF p4

Related: Ch 26.

- 7** A merchant has just purchased goods from a supplier based in an overseas country. The goods cost \$7 million. Currently these goods could be sold for \$10 million in the merchant's home country. It will be necessary to ship these goods from their present location and the voyage should take one month. The expenses of the voyage and selling the goods are estimated to be \$0.5 million.

The merchant estimates that over the duration of the voyage, one or more events could arise with the following total consequences and probabilities:

<i>Consequence</i>	<i>Probability of event</i>
Total loss of the cargo:	1%
Loss of half the value of the cargo:	2%
Loss of 5% of the value of the cargo:	40%

- (i) Calculate the insurance premium the merchant would expect to have to pay, in order to cover the risks of these events arising over the expected period of the voyage. [2]

The merchant decides not to take out this insurance cover.

It is estimated that, by the time the goods are sold at the end of the voyage, their sale value and the expenses incurred may change in the following ways:

Value of the goods:

No change:	60% chance
Increase by 20%:	25% chance
Decrease by 30%:	15% chance

Level of expenses:

No change:	40% chance
Increase by 25%:	40% chance
Decrease by 20%:	20% chance

- (ii) Calculate the expected profit the merchant will make on this venture. You should ignore tax and any interest payable or receivable. [5]

The merchant takes out a contract which guarantees that the goods will be sold to the writer of the contract in one month's time, for 97.5% of today's market value. At that time, the merchant must supply the goods to the writer of the contract at the merchant's home port, for which he will receive the agreed payment. The effect of taking out the contract is to reduce the expected profit to the merchant due to the cost of the guarantee.

- (iii) Discuss the possible financial implications for the merchant if the voyage takes longer than anticipated, for example at least an extra month. [8]
[Total 15]

CA1-P1 | April 2011 | Q1

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** A company provides a range of benefits to its employees.
- (i) Describe the key factors that influence the level and timing of contributions for these benefits. [3]
 - (ii) Outline why the expected cost of the benefit payments may be different from the price charged by an insurance company to insure these benefits. [5]
- [Total 8]

CA1-P1 | April 2012 | Q1

Current topic | 6 marks total | Original PDF p2

Related: Ch 20.

- 1** An insurance company offers two related life assurance policies.

The first policy is a standard five-year term assurance.

The second is a special five-year term assurance with an option at the end of the five year term for the policyholder to renew for a further five years without further underwriting. The premium rate at renewal will be set at the start of the special policy.

Discuss how the special initial and renewal premium rates applying under the second policy might differ from each other and from the standard premium rate for the first policy. [6]

CA1-P1 | April 2012 | Q7

Current topic | 22 marks total | Original PDF p6

Related: Ch 7, Ch 18.

- 7** The Wetsock music festival has been running for 25 years. The festival is open air and based in a remote location. Popularity has increased in the last five years with a large number of visitors choosing to camp at the site over the weekend. Although the acts at the festival are known in advance, late changes to scheduled performances are common.

There are two kinds of ticket and both require wristbands to be worn for the duration of the festival:

Silver band

Provides access to the festival and campsite.

Gold band

Provides access to the festival, campsite and VIP areas. In addition mobile text message alerts are provided when scheduled performances are confirmed or altered.

Next year, an insurance company will sponsor the festival. The insurance company has negotiated with the organisers to provide free insurance to gold band ticket holders. The insurance company will receive a share of the gold band ticket price in return for providing the insurance.

- (i) List the perils a visitor may wish to insure against during the festival weekend. [4]

The cover the insurance company decides to offer to gold band holders includes two key benefits:

- Mobile phone cover: a compensation payment if a visitor has their mobile phone stolen or damaged at the festival; and
- Cancellation cover: a full refund of the ticket price if the entire festival is cancelled due to poor weather.

- (ii) Discuss the factors that would need to be considered when determining the cost of providing the mobile phone cover at this particular festival. Your description should include possible sources of data. [8]

- (iii) Explain why the claims experience for the cancellation cover is likely to be markedly different and more difficult to predict compared to that for the mobile phone cover. [4]

- (iv) Outline the factors, which could lead to the price the insurance company would charge the organisers being different to the expected cost of claims. [6]
[Total 22]

CA1-P1 | September 2012 | Q3

Current topic | 11 marks total | Original PDF p2

Related: Ch 18.

- 3** A subsidiary of a multinational general insurance group is planning on launching a new car insurance product covering accident risks (i.e. not theft risk). This subsidiary writes business in only one small country and has no experience of writing car insurance risks.

The subsidiary has decided to use combined data from other companies in the multinational group to set the assumptions used to calculate premiums.

- (i) Describe how useful this data would be in determining these premium rate assumptions. [4]

The government of the subsidiary's country has banned the use of both age and gender as rating factors for all general insurance products.

- (ii) Discuss other rating factors that could be used to ensure that premiums charged by the subsidiary accurately reflect the risks associated with each policyholder. [7]
[Total 11]

CA1-P1 | April 2013 | Q6

Current topic | 20 marks total | Original PDF p3

Related: Ch 18, Ch 22.

- 6** An insurance company has introduced a new term assurance product. At outset, the policyholder selects an initial term of 10, 15, 20 or 30 years where the premium will be level, thereafter the policy continues without further underwriting with increasing yearly reviewable premiums. The policy ceases at age 95. The regulator requires that the initial policy documents specify guaranteed maximum premium rates for the yearly reviewable premiums.

- (i) Describe the sources of data that could be used to price the contract. [4]
- (ii) Explain why the regulator requires a guaranteed maximum premium. [2]
- (iii) Discuss how the insurance company might set the guaranteed maximum premium rates. [8]

After the insurance company has been selling the product for 12 years, they have decided to charge less than the guaranteed maximum premium rates.

- (iv) Suggest reasons why the insurance company has decided to charge rates lower than the maximum. [6]

[Total 20]

CA1-P1 | April 2014 | Q4

Current topic | 13 marks total | Original PDF p2

Related: Ch 17.

- 4** A government has invited several firms to tender for a construction contract. The project is to build a bridge, which has already been designed, across a river near a large city.

Government advisors have suggested that the bridge is expected to take between seven and fourteen months to construct based on a “standard workforce”, depending on working conditions. The expected outlay including material and labour is \$9 million.

The government will pay a fixed contract fee, to be determined during the tender process, if the bridge is constructed within twelve months. If the bridge is not completed within twelve months no fee will be paid.

The government has asked for the contract to be awarded to the firm that quotes the lowest contract fee, along with a suitable supporting project plan.

CA1-P1 | April 2014 | Q4 - continued

[Current topic](#) | 13 marks total | [Original PDF](#) p3

Related: Ch 17.

- (i) Describe how a construction firm could use actuarial techniques to assess whether to tender for this project and also to determine an appropriate contract fee to quote. [9]

In a further communication the government has confirmed that an early completion bonus is also available. The government will pay the bonus if the bridge is available for use ahead of the twelve month deadline. The bonus will be 5% of the original contract fee for each week ahead of the twelve month deadline.

- (ii) Discuss how the early completion bonus may affect the quote. [4]
[Total 13]

CA1-P1 | April 2016 | Q2

Current topic | 11 marks total | Original PDF p2

Related: Ch 30.

- 2** (i) List four factors that affect the risk of a claim on a motor insurance policy. [2]

A general insurance company writes a large amount of motor insurance business. The company is considering introducing a new motor insurance policy, where the insured driver will be required to fit a box to their vehicle to monitor their driving. This box will monitor how the insured is driving as well as where and when they drive.

- (ii) Outline the advantages of this approach. [3]

- (iii) Discuss how using this new method will affect the pricing of the product. [6]
[Total 11]

CA1-P1 | April 2018 | Q5

[Current topic](#) | [21 marks total](#) | [Original PDF p3](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 5** (i) State four methods a company could use to finance future benefits. [4]

An employer is setting up a benefit scheme for its employees.

- (ii) Explain the advantages and disadvantages of the different methods of financing benefits to the different parties involved with the benefit scheme. [10]

The employer is considering paying an insurance company to secure all of the schemes future benefits. The insurance company offers the employer the ability to pay for future benefits using one of the methods outlined in (i).

- (iii) Discuss the advantages and disadvantages of each method outlined in (i) for both parties. [7]

[Total 21]

CP1-P1 | April 2020 | Q4

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** An insurance company sells long-term care products in two countries. These countries are both developed and have similar population sizes.

The theoretical value of the benefits provided by the products is the same in both countries. The premiums charged are, however, very different.

Discuss possible reasons for the difference in premiums. [8]

CP1-P1 | September 2023 | Q4

Current topic | 12 marks total | Original PDF p2

Related: Ch 30, Ch 21.

4 Actpet is an insurance company that sells pet insurance.

- (i) Outline the factors that Actpet will need to consider when determining the premiums to charge for the pet insurance product. [6]

The market for pet insurance has become increasingly competitive and Actpet is investigating how the premiums they charge could be reduced.

- (ii) Describe how Actpet could control and manage the cost of the payments incurred and the company's administration expenses. [6]
- [Total 12]

CP1-P1 | September 2024 | Q9

Current topic | 12 marks total | Original PDF p4

Related: Ch 25, Ch 27.

9 An airline business transports cargo around the world.

- (i) List the main risks associated with the transportation of cargo that the airline is taking on. [3]

Currently the airline does not insure any of the risks identified in part (i).

- (ii) Describe how the airline could fund any significant losses arising from these risks. [4]

Following a year where the airline has had more losses than expected, the airline has decided to get quotations from general insurance companies for insurance cover for the transportation of cargo.

- (iii) Set out how a general insurance company would determine the premium for providing the cover. [5]

[Total 12]